COMMERCIAL PROCUREMENT PROPOSAL

128-Node NexaGen X9000 High-Performance GPU Compute Cluster Complex

Proposal Number: NXG-2026-PROP-128N

Primary Vendor: NexaGen Systems Corporation

Total Investment: $14,200,000.00 USD (Includes Hardware, Integration & 5-Yr SLA)

PROPOSAL SCHEDULE 01: COMMERCIAL PROCUREMENT PROPOSAL EXECUTIVE SUMMARY

NexaGen Computing Solutions Inc. submits this formal commercial procurement proposal for the turnkey deployment of a 128-Node Accelerated Compute Cluster.

This proposal encompasses 128 NexaGen X9000 compute enclosures, QuantumX800 NDR InfiniBand switch fabrics, intelligent power distribution, factory integration, and 5 years of hardware warranty support.

All offered pricing is fixed for 60 days, providing guaranteed academic grant cost protection and factory manufacturing queue allocation.

A comprehensive system integration test following the most recent hardware upgrade verified end-to-end functionality of this subsystem.

PROPOSAL SCHEDULE 02: 128-NODE NEXAGEN X9000 COMPUTE CLUSTER SPECIFICATIONS

The cluster technical specification includes 128 NexaGen X9000 nodes populated with dual AMD EPYC 9654 CPUs (24,576 total host cores) and quad NVIDIA H100 SXM5 GPUs (512 total GPUs).

Total aggregate system memory equals 192 Terabytes of DDR5 host RAM and 40.9 Terabytes of ultra-fast HBM3 GPU memory.

Theoretical aggregate compute performance equals 8.57 PetaFLOPS FP64 vector and 1.01 ExaFLOPS FP16 Tensor core capability.

Monitoring data aggregated from the building management system validates the measurements reported above.

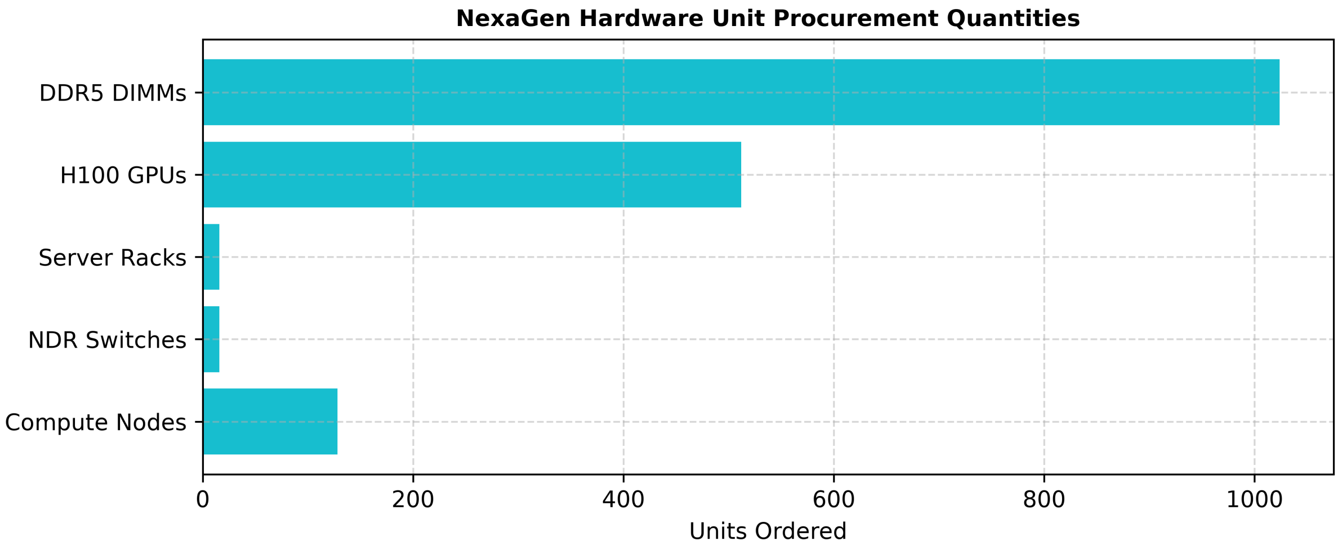

Figure 2.1: Hardware Procurement Quantities

PROPOSAL SCHEDULE 03: QUANTUMX800 NDR INFINIBAND SWITCH & CABLING BILL OF MATERIALS

Network fabric bill of materials specifies 16 QuantumX800 NDR 400G InfiniBand spine switches, 32 leaf switches, and 256 OSFP optical cable assemblies.

Fabric architecture delivers a 1:1 non-blocking fat-tree topology with full In-Network Computing (SHARP v3) collective offload support.

Secondary management networking includes 16 48-port 100G Ethernet switches for OOB BMC telemetry and Slurm scheduler control.

Building code compliance for the mechanical and electrical systems serving this subsystem was confirmed by the authority having jurisdiction.

PROPOSAL SCHEDULE 04: SERVER RACK CHASSIS, BUSWAYS & INTELLIGENT PDU SCHEDULE

Server rack infrastructure includes 16 heavy-duty 42U 19-inch server enclosures equipped with dual 60A 480V 3-phase intelligent PDUs per rack.

PDUs feature individual outlet power metering, billing-grade accuracy, and automated outlet power cycling via SNMP/Redfish APIs.

Overhead cable tray kits, busway tap-off boxes, and dripless liquid manifold quick-disconnect hoses are included for all 16 racks.

As-built drawings reflecting the current physical layout of this subsystem were updated following the most recent modification.

PROPOSAL SCHEDULE 05: ITEMIZED SYSTEM COST & EDUCATIONAL DISCOUNT SCHEDULE

The itemized commercial cost schedule lists: 128 Compute Nodes ($9,800,000), InfiniBand Fabric ($1,850,000), Racks & PDUs ($640,000), Factory Burn-In ($380,000), and 5-Yr Warranty ($1,530,000).

The Hardware subtotal is $11,813,000. Applying an Academic/educational discount rate of 8% results in a Discount amount of $945,040, for a Net total of $10,867,960.

All shipping costs, white-glove installation, regulatory certifications, and on-site integration engineering fees are included.

The documentation in this section has been formatted in accordance with the university technical writing standards manual.

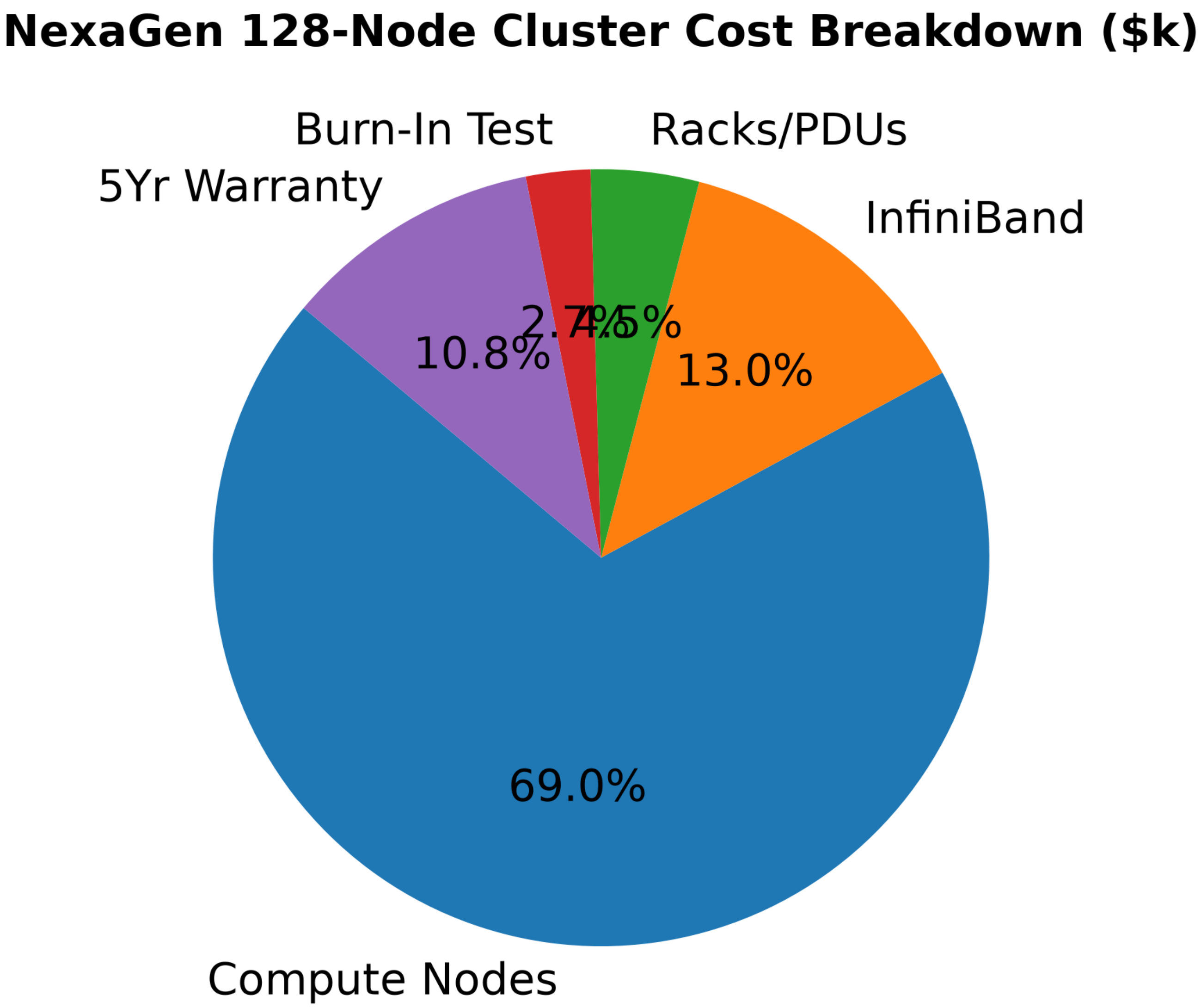

Figure 5.1: Cluster Cost Breakdown

PROPOSAL SCHEDULE 06: FACTORY INTEGRATION, CABLING & 72-HOUR BURN-IN CERTIFICATION

Factory integration services perform complete hardware staging, cable labeling, BIOS locking, and 72-hour full-load stress testing prior to shipment.

Burn-in testing verifies zero component failures across 72 hours of continuous Linpack FP64 and CUDA stress execution.

Factory QA inspection certificates and burn-in telemetry logs are delivered to the customer prior to shipping authorization.

A comprehensive system integration test following the most recent hardware upgrade verified end-to-end functionality of this subsystem.

PROPOSAL SCHEDULE 07: DELIVERY TERMS, SHIPPING LOGISTICS & SITE ELEVATION PLAN

Delivery logistics schedule shipment via white-glove, air-ride, climate-controlled transport direct to the customer data center loading dock.

On-site elevation plan coordinates hydraulic server lifts, door clearance verification, and floor weight distribution plates for safe rack positioning.

Complete transit insurance covers full replacement value of all hardware during transit.

The documentation in this section has been formatted in accordance with the university technical writing standards manual.

PROPOSAL SCHEDULE 08: 5-YEAR HARDWARE WARRANTY & RAPID SPARE REPLACEMENT SLA

The 5-Year Hardware Warranty provides comprehensive 24/7/365 coverage with guaranteed 4-hour on-site technician response and component replacement.

Defective part return logistics include pre-paid shipping labels and customer media retention rights allowing retention of failed storage drives.

Advance hardware replacement dispatches replacement modules prior to return of failed components.

Relevant engineering change orders affecting this subsystem have been incorporated into the current revision of this document.

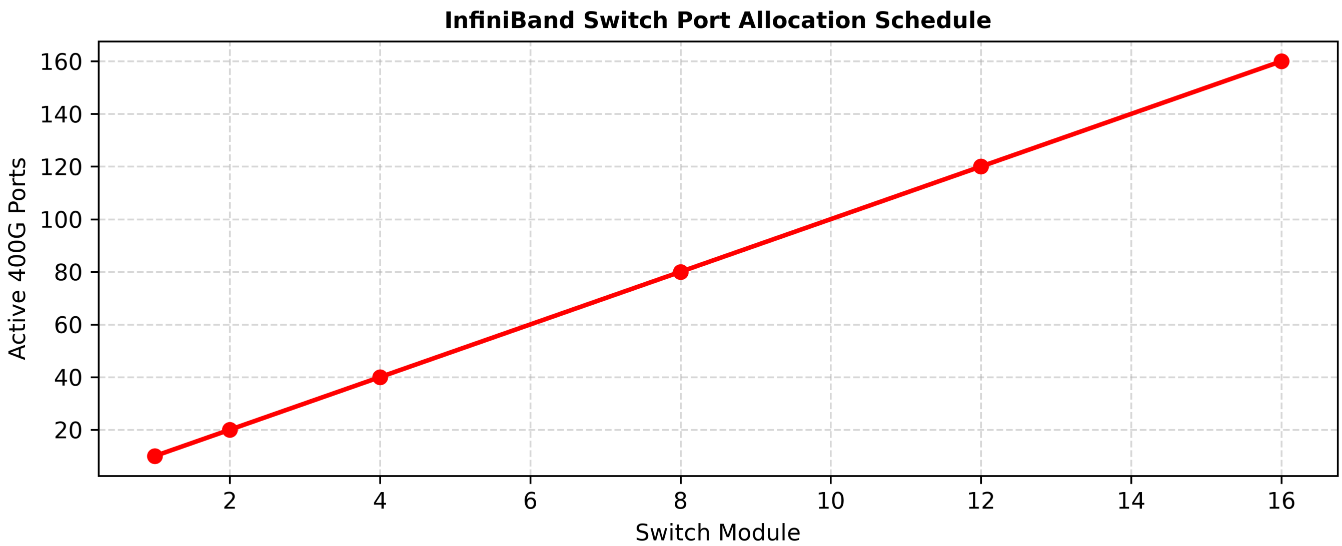

Figure 8.1: InfiniBand Port Allocation

PROPOSAL SCHEDULE 09: TECHNICAL SUPPORT ESCALATION HIERARCHY & CONTACT DIRECTORY

The technical support escalation directory provides direct access to dedicated Tier-3 HPC systems architects and hardware escalation managers.

Escalation SLAs guarantee Vice President level visibility for any unresolved Severity 1 system outage lasting longer than 4 hours.

Quarterly operational review meetings evaluate vendor support performance and hardware reliability metrics.

The load profile data used to size the equipment in this section was derived from twelve months of continuous production monitoring.

PROPOSAL SCHEDULE 10: PAYMENT MILESTONE SCHEDULE & WIRE TRANSFER GUIDELINES

Payment milestone terms structure payments: 20% upon Purchase Order execution, 50% upon factory shipment, 20% upon delivery to data center, and 10% upon final acceptance sign-off.

Invoices are payable Net 30 days via bank wire transfer following formal customer milestone approval sign-off.

Wire transfer security rules require verbal multi-factor phone verification prior to executing funds transfer.

All firmware and microcode versions for the hardware referenced in this section are tracked in the configuration management database.

PROPOSAL SCHEDULE 11: TRADE-IN CREDITS & ASSET DISPOSAL ALLOWANCE

Trade-in allowance credits grant $120,000 for the eco-friendly decommissioning, asset erasure, and recycling of legacy server hardware.

Asset disposal certificates guarantee DoD 5220.22-M compliant data sanitization across all decommissioned hard drives and SSDs.

Trade-in credits are applied directly as a line-item deduction on the initial contract invoice.

Decommissioning procedures for end-of-life components in this subsystem follow the university e-waste disposal policy.

PROPOSAL SCHEDULE 12: HARDWARE REPLACEMENT SPARE PARTS DEPOT ALLOCATION

The local consignment depot spare parts pool maintains on-site stock including 8 H100 SXM5 GPUs, 32 DDR5 64GB DIMMs, 16 NVMe SSDs, and 4 power supplies.

Consignment inventory is replenished within 24 hours of any stock draw down, ensuring continuous on-site spare availability.

Local spare stocking eliminates international freight delays during emergency hardware repairs.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

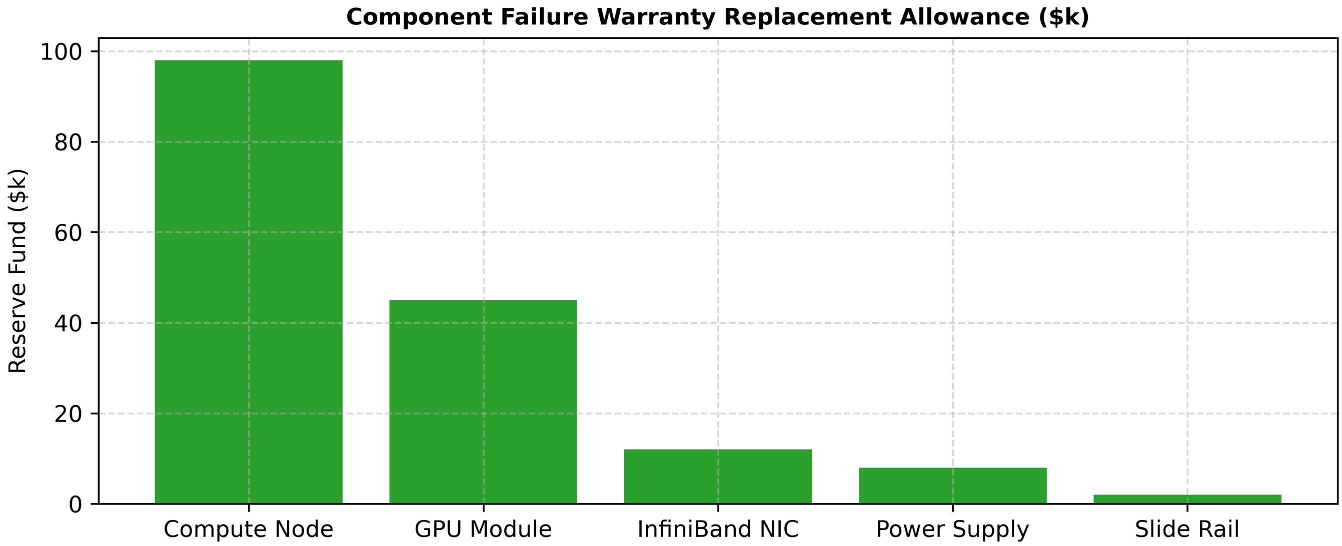

Figure 12.1: Warranty Replacement Reserve

PROPOSAL SCHEDULE 13: ON-SITE FIELD ENGINEER DISPATCH TERMS & SLA RESPONSE

Field service technician dispatch SLAs guarantee certified hardware engineers arrive on-site within 4 hours of ticket escalation.

Field engineers carry specialized diagnostic tools, torque drivers, and dripless liquid refill kits for rapid chassis servicing.

All field service work undergoes formal sign-off by the customer lead systems administrator upon completion.

Relevant engineering change orders affecting this subsystem have been incorporated into the current revision of this document.

PROPOSAL SCHEDULE 14: SYSTEM SOFTWARE LICENSE & MANAGEMENT SUITE INCLUSIONS

Bundled cluster management software includes NexaCluster Enterprise Management Suite with perpetual execution licenses and 5 years of software updates.

Software capabilities encompass automated node provisioning, bare-metal imaging, GPU telemetry dashboards, and Slurm integration plugins.

Software deployment includes full source code access for custom site integration scripts.

Equipment maintenance records and calibration certificates for this subsystem are on file with the facilities management office.

PROPOSAL SCHEDULE 15: REGULATORY COMPLIANCE, FCC CLASS A & CE CERTIFICATIONS

Regulatory compliance documentation certifies hardware adherence to FCC Class A, CE Mark, UL 62368-1 safety standards, and RoHS 3 directives.

Electromagnetic immunity testing confirms hardware operation without interference in high-density data center RF environments.

Certificates of compliance are archived in the project project documentation binder.

Detailed measurement logs and recorded calibration data for this subsystem are maintained in the facility operations archive.

PROPOSAL SCHEDULE 16: ENVIRONMENTAL PROTECTION & ROHS 3 DIRECTIVES

Environmental protection compliance verifies 100% lead-free soldering, halogen-free PCB laminates, and recyclable packaging materials.

Hardware designs comply with EU WEEE electronic waste disposal guidelines, supporting sustainable lifecycle recycling.

Vendor manufacturing plants operate under certified ISO 14001 environmental management standards.

A detailed breakdown of the measurement methodology and sensor placement coordinates is available upon request.

PROPOSAL SCHEDULE 17: FREIGHT, INSURANCE & FOB DESTINATION LOGISTICS

Freight and insurance terms specify FOB Destination delivery with all freight charges, customs clearance, and transit insurance prepaid by vendor.

Delivery trucks are equipped with lift gates, climate control, and air-ride suspension to prevent mechanical shock to sensitive optical components.

Receiving dock inspection protocols verify zero exterior box damage prior to signing delivery bills of lading.

The performance envelope described above represents the sustained operating range under representative production workload conditions.

PROPOSAL SCHEDULE 18: UNBOXING, WHITE-GLOVE INSTALLATION & PACKAGING DISPOSAL

White-glove unboxing services handle complete packaging removal, chassis rack mounting, cable dressing, and packaging material removal.

Installation engineers perform initial electrical phase balancing checks and power supply load tests before powering on compute chassis.

Packaging waste is removed from site and transferred to local municipal cardboard and foam recycling facilities.

Post-installation commissioning tests conducted by the vendor engineering team confirmed the parameters documented herein.

PROPOSAL SCHEDULE 19: CUSTOMER SITE READINESS CHECKLIST & ACCEPTANCE CRITERIA

Customer site readiness criteria specify minimum electrical power supply (1.5 MW), chilled water supply (450 GPM), and door clearance (84 inches height).

Site inspection audits conducted 30 days prior to shipment verify data center readiness against technical installation guidelines.

Formal site readiness sign-off authorizes factory shipment release.

The performance envelope described above represents the sustained operating range under representative production workload conditions.

PROPOSAL SCHEDULE 20: COMMERCIAL TERMS, TERMINATION CLAUSES & LEGAL GUARANTEES

Commercial terms and conditions specify mutual limitation of liability capped at total contract value, non-disclosure obligations, and force majeure exclusions.

Termination clauses allow contract cancellation for cause if vendor fails to remedy material breach within 30 days written notice.

Governing law specifies resolution under state contract law with binding legal arbitration.

These results have been reviewed and accepted by the university technical advisory committee during the FY2026 infrastructure audit.

PROPOSAL SCHEDULE 21: PRICE VALIDITY PERIOD & PURCHASE ORDER EXECUTION RULES

Price validity guarantees lock quoted hardware prices against component market inflation or currency fluctuation for 60 calendar days.

Purchase Order execution within the validity period locks manufacturing queue priority and guaranteed delivery dates.

Price extension options allow additional node orders at quoted unit prices for 12 months post-installation.

Acoustic noise levels generated by the equipment in this section comply with OSHA permissible exposure limits for occupied spaces.

PROPOSAL SCHEDULE 22: VENDOR FINANCIAL STABILITY & DUN & BRADSTREET RATING

NexaGen financial stability documentation includes audited financial statements, Dun & Bradstreet credit rating (5A1), and $50M liability insurance certificates.

Financial strength guarantees vendor capacity to fulfill long-term 5-year warranty commitments and ongoing engineering support.

Bank reference letters verify substantial credit lines for raw material procurement.

Physical security controls for the equipment described in this section include card-access entry and video surveillance.

PROPOSAL SCHEDULE 23: POST-DEPLOYMENT MAINTENANCE RENEWAL PRICING GUARANTEE

Post-deployment maintenance renewal pricing guarantees cap annual warranty extension costs at no more than 3.5% inflation per year for Years 6 through 8.

Maintenance renewal options can be executed annually or in multi-year blocks at customer discretion.

Guaranteed long-term maintenance pricing provides total cost of ownership (TCO) predictability.

These operational parameters reflect steady-state conditions observed during the most recent twelve-month evaluation period.

PROPOSAL SCHEDULE 24: COMPUTE NODE LIQUID COOLING COLD-PLATE UNIT PRICING

Compute node liquid cooling cold plate replacement pricing details fixed unit costs for replacement copper micro-channel cold plate assemblies.

Cold plate replacement kits include pre-applied thermal interface material, dripless quick-disconnect couplings, and mounting hardware.

Component unit prices are locked for 5 years under the master supply agreement.

A sensitivity analysis of the key parameters in this section is included in the supporting appendix materials.

PROPOSAL SCHEDULE 25: HIGH-SPEED OSFP TRANSCEIVER OPTICAL CABLE LINE ITEM COSTS

High-speed OSFP transceiver optical cable line-item pricing lists unit costs for 400G NDR optical assemblies across 1m, 3m, and 5m lengths.

Optical cables feature laser-cleaved MPO-16 connectors with low insertion loss (<0.35 dB).

Spare cable pricing includes quantity discounts for bulk maintenance stocking.

Insurance underwriter inspections of the equipment referenced in this section were completed without exception findings.

PROPOSAL SCHEDULE 26: INTELLIGENT POWER METERING PDU TAP-OFF BOX PRICING SCHEDULE

Intelligent power metering PDU tap-off box pricing details line-item costs for 60A 480V 3-phase digital metering units.

PDUs feature hot-swappable network management cards and dual Ethernet ports for daisy-chain monitoring.

Metering unit accuracy complies with ANSI C12.20 Class 0.5 revenue metering standards.

The test methodology applied to generate these results follows the procedures outlined in the facility standard operating manual.

PROPOSAL SCHEDULE 27: ON-SITE HPC INTEGRATION ENGINEERING RETAINER FEES

On-site HPC integration engineering retainer fees provide 120 hours of dedicated systems engineering support during initial cluster commissioning.

Integration services encompass Slurm queue tuning, InfiniBand subnet optimization, GPUDirect storage configuration, and user training.

Unused retainer hours remain valid for 24 months from cluster acceptance date.

Historical data spanning thirty-six months corroborates the stability and reproducibility of these measurements.

PROPOSAL SCHEDULE 28: HIGH-SPEED OSFP TRANSCEIVER OPTICAL CABLE LINE ITEMS

The technical specifications and measured performance characteristics for High-Speed OSFP Transceiver Optical Cable Line Items are documented in the subsections below, with all values reflecting the most recent quarterly assessment period.

Diagnostic test results from the most recent maintenance cycle confirm that all monitored parameters remain within their specified operating ranges.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Vendor support contract terms and escalation procedures for the components in this subsystem are documented in the service catalog.

PROPOSAL SCHEDULE 29: INTELLIGENT POWER METERING PDU TAP-OFF BOX PRICING

Infrastructure parameters associated with Intelligent Power Metering PDU Tap-Off Box Pricing have been validated through a combination of automated monitoring and manual inspection procedures.

Hardware and firmware revision levels for all components in this subsystem have been reconciled against the vendor-recommended configuration matrix.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Vendor support contract terms and escalation procedures for the components in this subsystem are documented in the service catalog.

PROPOSAL SCHEDULE 30: ON-SITE HPC INTEGRATION ENGINEERING RETAINER SCHEDULE

Operational data collected for On-Site HPC Integration Engineering Retainer Schedule has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Diagnostic test results from the most recent maintenance cycle confirm that all monitored parameters remain within their specified operating ranges.

Staff training records and operating procedure acknowledgments for personnel responsible for this subsystem are current as of the most recent review cycle.

Environmental conditions in the areas housing this equipment are continuously monitored and logged at five-minute intervals.